

Advanced Statistical Methods II

Practice Problems on LAD Regression

1 Setup and Notations

- Data: $(x_1, y_1), \dots, (x_n, y_n)$, where y is the response and x is the predictor.
- **LAD with intercept.** Minimise

$$Q(a, b) = \sum_{i=1}^n |y_i - a - bx_i|, \quad r_i = r_i(a, b) = y_i - a - bx_i.$$

- **LAD without intercept.** Minimise $Q_0(b) = \sum_{i=1}^n |y_i - bx_i|$.
- **OLS** replaces $|\cdot|$ by $(\cdot)^2$. Its normal equations (with intercept) are $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$, where e_i are the OLS residuals.
- Throughout this document, assume that the x_i 's are *not all equal*.
- **Population version:** $Q(a, b) = E|Y - a - bX|$, assuming $E|Y| < \infty$.
- n_+, n_-, n_0 denote the numbers of positive, negative and zero residuals, respectively.

2 Useful Results

The following results would be useful to solve the problems.

Tool A (KKT conditions). (a, b) minimises Q if and only if there exist numbers $s_1, \dots, s_n$ with

$$s_i = \text{sign}(r_i) \text{ if } r_i \neq 0, \quad s_i \in [-1, 1] \text{ if } r_i = 0,$$

such that

$$\sum_{i=1}^n s_i = 0 \quad (\text{intercept equation}), \quad \sum_{i=1}^n s_i x_i = 0 \quad (\text{slope equation}).$$

Without intercept, only the slope equation is needed. We shall be referring to such (s_i) as a *certificate*.

Tool B (Uniqueness). Suppose (s_i) is a certificate at (a, b) , and there are two points j, k with $r_j = r_k = 0$, $x_j \neq x_k$, and $|s_j| < 1$, $|s_k| < 1$. Then (a, b) is the *unique* LAD solution.

Tool C (Linear conditional medians). If $m(x) = a_0 + b_0 x$ is a conditional median of Y given $X = x$ (for almost every x) and $E|Y| < \infty$, then (a_0, b_0) minimises $E|Y - a - bX|$. (Analogously, if $E(Y | X) = a_0 + b_0 X$ and $EY^2 < \infty$, then (a_0, b_0) is the population OLS solution.)

Tool D (Weighted median). Let $w_i > 0$, $W(A) = \sum_{i \in A} w_i$ and $W = \sum_i w_i$. Then b minimises $g(b) = \sum_i w_i |z_i - b|$ if and only if

$$W(z < b) \leq W/2 \leq W(z \leq b),$$

where $W(z < b) = \sum_{i: z_i < b} w_i$ and $W(z \leq b) = \sum_{i: z_i \leq b} w_i$.

Try to prove these four tools and proceed with the following problems.

3 Practice Problems

Problem 1. Suppose not all x_i 's are equal. Consider LAD regression of y on x with intercept.

- (a) Show that a minimiser of $Q(a, b)$ exists.
- (b) Show that there is *some* LAD line which passes through at least two data points (x_j, y_j) and (x_k, y_k) with $x_j \neq x_k$.
- (c) Must *every* LAD line pass through two data points?

Problem 2. Two disjoint subpopulations S_1 and S_2 are given. We fit LAD with intercept of y on x separately in each. Suppose both fits give the same optimal coefficients (α^*, β^*) . Now regress y on x by LAD on $S_1 \cup S_2$.

- (a) Is (α^*, β^*) an LAD solution for the union?
- (b) Must the LAD estimates for the union be (α^*, β^*) ?
- (c) What if we consider the same question for OLS?

Problem 3. Consider the same setup as in Problem 2. Now suppose that in the two subpopulations the LAD fits are

$$(\alpha, \beta_1) \quad \text{and} \quad (\alpha, \beta_2), \quad \beta_1 \neq \beta_2, \quad \beta_1, \beta_2 > 0,$$

i.e. the same intercept but different slopes. We fit LAD on the union.

- (a) Does the intercept remain equal to α ?
- (b) Does the pooled slope lie between β_1 and β_2 ?

Problem 4. Consider the same setup as in Problem 2, but now the subpopulation LAD fits are (α_1, β_1) and (α_2, β_2) with

$$0 < \alpha_1 < \alpha_2, \quad 0 < \beta_1 < \beta_2.$$

- (a) Must the pooled LAD estimates satisfy $\hat{\alpha} \in [\alpha_1, \alpha_2]$ and $\hat{\beta} \in [\beta_1, \beta_2]$?
- (b) Must $\hat{\beta}$ at least be positive?
- (c) Is there a natural condition under which the pooled estimates *are* in the intervals?

Problem 5. We fit no-intercept OLS and LAD regressions of y on x and observe that the two slope estimates are the same: $\hat{b}_{OLS} = \hat{b}_{LAD}$.

- (a) Characterise exactly when this happens.
- (b) Does it imply that the residuals are symmetric?
- (c) Does it imply that the through-origin model is correct?

Problem 6. We fit OLS and LAD regressions of y on x with intercept, and observe that both the intercepts and the slopes coincide.

- (a) What does this tell us about the residuals?
- (b) Does it imply symmetric errors?
- (c) In the location-scale model $Y = \alpha + \beta X + (\gamma + \delta X)\varepsilon$, when are the OLS and LAD targets equal?
- (d) How likely is exact equality in a sample?

Problem 7.

- (a) Show that LAD with only an intercept gives the sample median.
- (b) Show that LAD with intercept and a binary predictor $x \in \{0, 1\}$ gives the two group medians.
- (c) Show that no-intercept LAD gives an $|x|$ -weighted median of y_i/x_i .
- (d) Compute the no-intercept LAD and OLS estimates for the data

$$(1, 1), \quad (2, 5), \quad (3, 3), \quad (4, 9).$$

Problem 8. Let $(\hat{a}, \hat{b})$ be any LAD fit with intercept.

- (a) Show that $n_- \leq n/2 \leq n_- + n_0$ (equivalently, $n_+ \leq n/2$ and $n_- \leq n/2$).
- (b) For quantile regression at level $\tau \in (0, 1)$, with loss $\rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\})$, show that $n_- \leq n\tau \leq n_- + n_0$.
- (c) Converse: if a line has $n_+ \leq n/2$ and $n_- \leq n/2$, must it be an LAD line?

Problem 9.

- (a) For the unit-square data $(0, 0), (0, 1), (1, 0), (1, 1)$, find the entire set of LAD solutions (a, b) .
- (b) Relate the extreme points of this set to Problem 1.
- (c) Give a simple sufficient condition for uniqueness.

Problem 10. Let $(\hat{a}, \hat{b})$ be an LAD fit.

- (a) Move any y_i vertically without changing the sign of its residual (points on the fitted line stay put). Show that $(\hat{a}, \hat{b})$ remains an LAD fit. What happens to OLS?
- (b) Show that a single point with extreme x can make the LAD slope any real number.

Problem 11. Let $(\hat{a}, \hat{b})$ be an LAD fit, with fitted value $\hat{y}_0 = \hat{a} + \hat{b}x_0$ at some x_0 .

- (a) Add a point on the fitted line, $(x_0, \hat{y}_0)$. Does the fit change?
- (b) Add a symmetric pair $(x_0, \hat{y}_0 + d)$ and $(x_0, \hat{y}_0 - d)$, $d > 0$. Does the fit change?
- (c) Can adding a single point off the line change the fit?

Problem 12. Let $g_i \in \{0, 1\}$ indicate the subpopulation. Fit by LAD the full-interaction model

$$y = a + bx + c g + d (g \cdot x).$$

- (a) Show that this reproduces the two separate subpopulation LAD fits.
- (b) Contrast this with pooling (Problems 3 and 4).

Problem 13. Let $(\hat{a}, \hat{b})$ be an LAD fit of y on x (with intercept).

- (a) Find the LAD fit of $y^* = cy + a_0 + b_0x$ on x , where $c \neq 0$.
- (b) Find the LAD fit of y on $x^* = px + q$, where $p \neq 0$.
- (c) Population: if $\text{med}(\log Y \mid X) = a + bX$, what is $\text{med}(Y \mid X)$? Is the analogous statement true for conditional means?
- (d) Is $\text{med}(h(Y)) = h(\text{med}(Y))$ for every function h ?

Problem 14.

- (a) Let $F = \pi F_1 + (1 - \pi)F_2$ with $\pi \in (0, 1)$, and let $m_1 \leq m_2$ be medians of F_1 and F_2 respectively. Show that F has a median in $[m_1, m_2]$.
- (b) In the two-group model (group independent of X), each group has a linear conditional median. Must the pooled conditional median $\text{med}(Y \mid X = x)$ be linear in x ? Must it be the π -weighted average of the group medians?

Problem 15.

- (a) Construct two subpopulations with the same positive LAD slope for which the pooled LAD slope is negative.
- (b) Show that adding a group indicator (common slope, separate intercepts) recovers the correct slope in your example.